

Evaluating Spectral Interpretability in Brazilian Portuguese Audio Deepfakes via Causal Occlusion

Como as pistas acústicas de um detector XLS-R mudam entre zero-shot e adaptação leve — TODO:
confirmar/traduzir

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ABSTRACT

TODO: abstract.

KEYWORDS

audio deepfake detection, anti-spoofing, explainability, SHAP, spectral occlusion, MFCC, cross-lingual, wav2vec 2.0, XLS-R, Brazilian Portuguese

1 INTRODUCTION

TODO.

We frame the study around one research question: *which acoustic cues does the detector rely on, and how do these cues shift after a light adaptation to Brazilian Portuguese?* We evaluate two detectors sharing a frozen XLS-R front-end: a zero-shot detector (D_{zs}) and a lightly adapted one (D_{ad}).

Two conditions must hold for the comparison to be meaningful. First, adaptation must improve detection: we require $EER(D_{ad}) < EER(D_{zs})$ before any cue analysis; otherwise the “before/after” contrast is reported as *insufficient adaptation*. Second, the acoustic surrogate must be faithful: SHAP-based cue attributions are interpreted only where the MFCC surrogate reproduces the detector score with high fidelity (R^2 and rank correlation on a held-out split); low fidelity is itself reported as a finding.

We investigate three hypotheses:

- **H1 (associative cue shift).** The acoustic-importance profile over MFCC features differs between D_{zs} and D_{ad} . We assess H1 *descriptively*, by comparing per-feature SHAP importances; we make no significance claim here.
- **H2 (causal spectral shift).** The frequency bands whose occlusion reduces $P(\text{spoof})$ differ between D_{zs} and D_{ad} . Because occlusion is an intervention on the input, we describe the resulting causal profiles, again *descriptively*, without a significance test.
- **H3 (error-linked acoustic differences).** For the MFCC features most salient under SHAP, distributions differ across decision quadrants: in mean between true-negative and false-positive clips (Welch’s t -test) and in variance between true-positive and false-negative clips (Levene’s test). H3 is tested *inferentially*, controlling the false discovery rate (Benjamini–Hochberg, $q < 0.05$).

2 RELATED WORK

2.1 Self-Supervised Front-Ends for Anti-Spoofing

Tak et al. [4] TODO.

2.2 Explainability in Audio Deepfake Detection

Grinberg et al. [3] and Fullwood and Monroe [2] TODO.

2.3 Cross-Lingual Generalization and Real-World Deployment

Channing et al. [1] TODO.

3 METHODOLOGY

3.1 Detectors Under Study

TODO.

3.2 Pre-processing

TODO.

3.3 Acoustic Features (MFCC)

TODO.

3.4 Spine 1 — Surrogate + SHAP

TODO.

3.5 Spine 2 — Spectral Causal Occlusion

TODO.

3.6 Confirmatory Statistics

TODO.

4 EXPERIMENTAL SETUP

5 RESULTS

5.1 Zero-Shot vs. Adapted Performance

TODO.

6 EXPLAINABILITY ANALYSIS

6.1 Acoustic Importance via SHAP

TODO.

Table 1: Experimental configuration.

Parameter	Value
Sample rate	16,000 Hz
Fixed window	64,600 samples (≈ 4.04 s)
Front-end	wav2vec 2.0 / XLS-R (frozen)
Checkpoint	nii-yamagishilab/mms-300m-anti-deepfake
Adaptation head (D_{ad})	logistic regression on 1024-d embeddings
Surrogate model	boosted trees $\rightarrow$ TreeSHAP
MFCC features	13 coeff. $\times$ (mean, std) = 26
Dataset	AKCIT-Deepfake/BRSpeech-DF
Train per class (D_{ad} head)	1,500 (balanced)
Analysis set per class (test)	1,500 (balanced)
Positive class	spoof
Seed	42

[2] Boo Fullwood and Fabian Monrose. 2025. Seeing is Believing: Interpreting Behavioral Changes in Audio Deepfake Detectors Arising from Data Augmentation. In *Proceedings of the 2025 Workshop on Artificial Intelligence and Security (AISec '25)*. ACM, Taipei, Taiwan, 206–217.

[3] Petr Grinberg, Ankur Kumar, Surya Koppiseti, and Gaurav Bharaj. 2025. What Does an Audio Deepfake Detector Focus on? A Study in the Time Domain. In *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 1–5.

[4] Hemlata Tak, Massimiliano Todisco, Xin Wang, Jee-weon Jung, Junichi Yamagishi, and Nicholas Evans. 2022. Automatic Speaker Verification Spoofing and Deepfake Detection Using Wav2vec 2.0 and Data Augmentation. In *Proc. The Speaker and Language Recognition Workshop (Odyssey 2022)*. 112–119. <https://doi.org/10.21437/Odyssey.2022-16>

Table 2: Master table schema (one row per audio clip; `_zs` = zero-shot, `_ad` = adapted).

Column	Description
sample_id	unique clip identifier
ground_truth	0 = bonafide, 1 = spoof
spoof_algorithm	generator/attack (null if absent)
p_spoof_{zs,ad}	continuous P (spoof) per detector
pred_{zs,ad}	decision at EER threshold
quadrant_{zs,ad}	TP / TN / FP / FN
mfcc{i}_mean	mean of i -th MFCC ($i = 1..13$)
mfcc{i}_std	std of i -th MFCC ($i = 1..13$)

Table 3: Detection performance and surrogate fidelity (performance_table).

Detector	EER	MCC	Accuracy	Surrogate R^2
D_{zs} (zero-shot)	TODO	TODO	TODO	TODO
D_{ad} (adapted)	TODO	TODO	TODO	TODO

6.2 Spectral Occlusion Profiles

TODO.

6.3 Confirmatory Tests

TODO.

7 DISCUSSION AND LIMITATIONS

TODO.

8 CONCLUSION AND FUTURE WORK

TODO.

REFERENCES

[1] Georgia Channing, Juil Sock, Ronald Clark, Philip Torr, and Christian Schroeder de Witt. 2024. Toward Robust Real-World Audio Deepfake Detection: Closing the Explainability Gap. *arXiv preprint arXiv:2410.07436* (2024).